Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to safely test a predicted circuit result.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using specific numbers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ammeter” and “voltmeter” to explain your answer. Sentence starters are optional.

ammeter: an instrument placed in series to measure current

voltmeter: an instrument placed across two points to measure voltage

trial: one run or repetition of a test

01 Warm-up: bring back an earlier idea

An ideal series loop has supply **15 V**. One component drops **3 V**. **Find the other component’s drop**.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

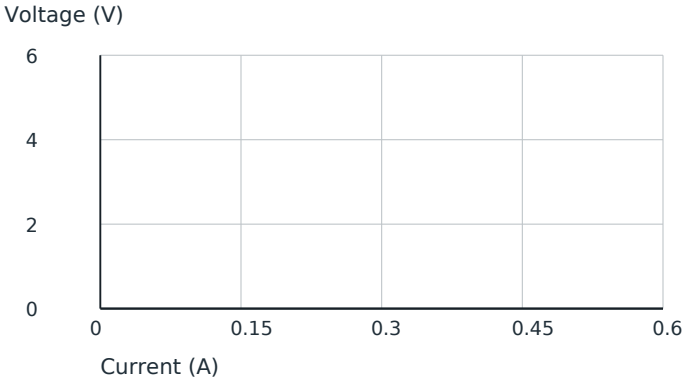
02 Worked example: explain the prediction

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=**2**, **4**, **6 V** and I=**0.2**, **0.4**, **0.6 A**. Each V/I is **10 ohms**. This supports treating the resistance as constant over the measured range.

Your turn: Plot V against I and **mark the slope**. **State why the data** do not show that the same relationship holds at every possible temperature or voltage.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On a resistor circuit sketch, add an ammeter to **measure loop current** and a voltmeter to **measure the resistor’s drop**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

Before powering a teacher-approved circuit, predict current for a **6 V** supply and **12 Ω** resistor. Then **use the table** to **record actual voltage**, current and observations during the test. Keep measured entries blank until you take measurements.

Sentence starter: I predict … . I used … to make this prediction because … .

Voltage (V)	Current (A)	Observation

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A reading disagrees with your prediction. **List two checks before changing** your explanation or deleting measurements.

Sentence starter: I predict … . I used … to make this prediction because … .

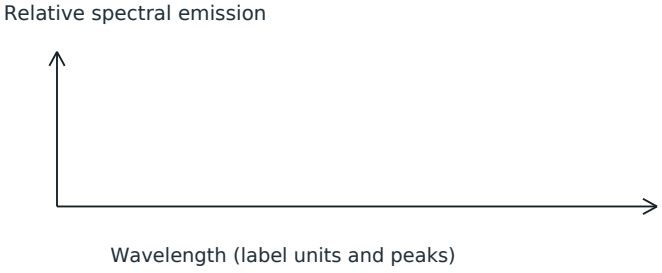
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An ideal blackbody's kelvin temperature becomes **4** times as large. How does its peak wavelength change? **Show how the thermal peak** changes.

Word help: blackbody = an ideal absorber/emitter used in a thermal-radiation model

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

07 Exit check

Before the first powered test, have the teacher check the circuit connections and meter placement. **Mark the checked locations** on the diagram and **record one actual comment or** change. Do not invent feedback.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A later reading is **8 V** and **0.5 A**. Does it fit the same **10 ohm** resistance? **Name two checks before declaring** the physical law wrong.

Sentence starter: The step I would change is … . I would change it because I found … .

Quantity	Lower bound	Central value	Upper bound
Voltage (V)	____	____	____
Current (A)	____	____	____
Resistance (Ω)	____	____	____

Marzano check - circle: 1 2 3